rules.md

> **CivicSync · Business Rules** Version 1.0-rc1 · 28 Oct 2026 · Owner: Backend Lead (Manindra Shukla) · Co-owner: UX & Ops Lead (Somil Tomar) **This is the single source of truth for every number that can be tuned.** Other documents point here. The values are **pilot defaults** (**A-05**, **A-07**): they are our best starting point and are expected to be tuned in Sprint 5 with a ward officer's input (**Q-03**). In the database they live in `rule_config` (§9).

1. Triage scoring

1.1 The formula

```
PriorityScore = round( 100 × ( 0.30·C + 0.25·E + 0.15·K + 0.30·L ) ) + B
capped at 100
```

Symbol	Meaning	Where it comes from
C	Class weight (how dangerous this kind of hazard usually is)	§1.2
E	Extent (how big it looks)	§1.3
K	Model confidence, 0–1 (how sure the phone's guess was)	From the report. If the person *changed* the class, K is set to 1.0 for the person's choice and the model confidence is kept separately.
L	Location context (is it somewhere many people or vulnerable people pass?)	§1.4
B	Corroboration bonus (several different people reported it)	§1.6

Weights sum to 1.00 (0.30 + 0.25 + 0.15 + 0.30). They are in `rule_config` as `score.w_class`, `score.w_extent`, `score.w_conf`, `score.w_location`.

1.2 Class weight C

Class	C
`open_drain` (open drain or missing cover)	1.00
`pothole`	0.80
`broken_road`	0.60
`waterlogging`	0.50
`other`	0.40

Why: an open drain or missing cover is a fall hazard with serious injury potential; potholes are the biggest source of two-wheeler crashes; waterlogging is often temporary but can hide potholes, so it is weighted lower and re-checked after rain.

1.3 Extent E

Size chosen	E
Large	1.00
Medium	0.60
Small	0.30
Not sure	0.50

1.4 Location context L

Take the **highest** value that applies.

Condition	L
Within 200 m of a school or hospital	1.00
Within 30 m of a primary or secondary road (including highway approaches)	0.90
Within 200 m of a market, bus stand or railway station	0.70
Local or residential street, nothing above applies	0.40
Unknown (not yet looked up)	0.50

1.5 Preliminary severity on the phone

The phone uses the **same formula** with L = 0.50 (unknown) and B = 0, and sends the result as `severity_prelim`. The cloud ignores it for ranking and recomputes with the true L (**D-10**). The phone value is kept so we can measure how far the two drift apart.

1.6 Corroboration bonus B

```
B = min(10, 3 × (distinct_reporters − 1))
```

Distinct reporters means different accounts. Several reports from the same account within 24 hours at the same spot count once (§6.3).

1.7 Bands

Band	Score	Citizen never sees this word; officers do
Critical	75–100	
High	55–74	
Medium	35–54	
Low	0–34	

1.8 Worked examples

Case	C	E	K	L	Score
Large pothole, main road, model 0.90 sure, 1 reporter	0.80	1.00	0.90	0.90	100 × (0.24 + 0.25 + 0.135 + 0.27) = 89.5 → **90, Critical**
Medium waterlogging, residential lane, 0.70 sure, 1 reporter	0.50	0.60	0.70	0.40	100 × (0.15 + 0.15 + 0.105 + 0.12) = 52.5 → **53, Medium**
Same waterlogging, now 4 distinct reporters					53 + min(10, 9) = **62, High**
Small broken road, unsure size not given, near a school, 0.80 sure	0.60	0.30	0.80	1.00	100 × (0.18 + 0.075 + 0.12 + 0.30) = 67.5 → **68, High**

1.9 Ranking order in the queue

1. Band (Critical first). 2. Score, highest first. 3. Oldest `first_reported_at` first.

1.10 "Why this rank?" text

The dashboard builds a sentence from the biggest contributors, e.g. *"High priority: large pothole on a main road, reported by 3 people."* The sentence template is in the app's string files (`design.md` §5.2).

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2. Report intake and quality rules

#	Rule	Value	What happens
2.1	Location accuracy good	`gps_accuracy_m` ⩽ 30 m	Normal.
2.2	Location accuracy weak	> 30 m and ⩽ 100 m	App asks the person to nudge the pin; report flagged `low_gps`.
2.3	Location accuracy very weak	> 100 m	App requires the person to place the pin manually; flagged `low_gps`.
2.4	Outside coverage	Point not inside any pilot ward polygon	Report is not saved. Person sees the "outside pilot area" message (`design.md` §7.2).
2.5	Stale capture	Captured more than 72 hours before it reaches the server	Accepted but flagged `stale_capture` for the officer.
2.6	Mock location	Phone says location is simulated	Accepted but flagged `mock_location`; excluded from auto-verification (§3.2).
2.7	Exact copy photo	Same SHA-256 already in the system	Not saved twice. Person sees: "This photo has already been sent."
2.8	Blurry photo	Sharpness check fails	Person is offered a retake; they can choose "Use anyway"; flagged `blurry`.
2.9	Rate limit	10 reports per account per day	After that, the kind message in `design.md` §7.2. Limit is in `rule_config`.
2.10	Age	Self-declared 18+ at first launch in the pilot (**A-10**)	Under-18 users can't continue.
2.11	Note length	⩽ 200 characters	
2.12	Privacy blur failed	Face or plate detection errored	Report is sent but flagged `privacy_blur_failed`; photo visible to officers

#	Rule	Value	What happens
			only, and the retention clock for such photos is halved (§7).

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3. Verification rules

3.1 Who can verify

A **Ward Officer** (for their wards), a **Zonal Engineer** or a **City Admin**. Crew Leads and citizens can't verify.

3.2 Verification by corroboration

A `reported` ticket becomes `verified` automatically when **all** of these hold:

- It has **3 or more distinct reporters** within the last 14 days, and
- at least **2** of their reports have no `mock_location`, `blurry` or `stale_capture` flag, and
- all counted reports are within 25 m of the ticket's point.

The timeline shows "Checked (by several reports)" so it's clear no officer has looked yet. An officer may still reject it.

3.3 Fast lane (not auto-verification)

A single report with model confidence ⩾ 0.85, class confirmed unchanged by the person, `gps_accuracy_m` ⩽ 20 m, no quality flags, and inside a ward gets a **Fast lane** badge. It still needs an officer's click, but sits in a "quick confirm" strip at the top of the queue. This saves time without letting a model verify alone.

3.4 Rejection reasons (fixed list)

`not_a_road_hazard`, `unclear_photo`, `outside_area`, `duplicate_merged`, `test_or_misuse`. A reason is required. Citizen wording is in `design.md` §7.3.

3.5 Assignment and resolution

- Assign only to a Crew Lead whose wards include the ticket's ward.
- Resolve requires an **after-photo**. If the photo has a location and is more than 50 m from the ticket, it's accepted but flagged `after_photo_far` for the Ward Officer.
- An officer can unassign (back to `verified`) with a note.

3.6 "Still there?" check and reopening

- When a ticket becomes `resolved`, the original reporters get a gentle push: "Marked as fixed. Is it really?"

- **Two different people** saying "Still there" within **14 days** reopen the ticket to `verified` and add 1 to `reopen_count`. A Ward Officer can also reopen at any time.
- A ticket reopened **twice** is escalated straight to level 2 (§5).

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4. SLA rules

An SLA clock runs for the **current stage** only. Defaults (**A-07**), to be agreed with the pilot municipal partner:

Band	Verify (from `reported`)	Assign (from `verified`)	Resolve (from `assigned`)
Critical	4 working hours	1 working day	3 working days
High	1 working day	2 working days	7 working days
Medium	2 working days	5 working days	14 working days
Low	3 working days	7 working days	30 working days

- **Working time:** Monday–Saturday, 10:00–17:00 IST, excluding a holiday list kept in `rule_config`.
- **Band changes:** if a ticket's band goes up (for example, more reporters join), its current stage deadline is recalculated from the stage start using the faster band. It never gets *longer* because of a band change.
- **Pause:** an officer can pause a clock for one of: `waiting_for_materials`, `heavy_rain`, `needs_other_department`. A pause is capped at 7 days, needs a note, and is visible on the timeline.
- **Citizens are not shown SLA numbers.** They see honest statuses and, if something is late, "This is taking longer than we hoped. We've asked for an update."

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5. Escalation rules

Level	Trigger	Who is alerted	How
L1	Stage clock reaches 100%	The person responsible for the stage (Ward Officer for verify/assign, Crew Lead for resolve)	Dashboard badge and email
L2	150%	Zonal Engineer	Dashboard Overdue list and email
L3	200%	City Admin	Weekly overdue summary and dashboard flag
Instant L2	Ticket reopened twice (§3.6)	Zonal Engineer	As above

An escalation never changes the Priority Score. It's recorded as a `ticket_events` row of type `escalated` and `escalation_level` is updated. A scheduled job (`pg_cron`) checks every 15 minutes.

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6. Duplicate thresholds

6.1 Compatibility groups

Reports can only merge if their classes are in the same group:

Group	Classes
Surface damage	`pothole`, `broken_road`
Drain	`open_drain`
Water	`waterlogging`
Other	`other` (merge only if an officer approves)

6.2 Distance thresholds

Distance to nearest compatible open ticket	Action
⩽ **25 m**, and both points have accuracy ⩽ 30 m	**Auto-merge.** Report joins the ticket.
⩽ 25 m but either point has accuracy > 30 m	Create a new ticket and add a **candidate** for officer review.
> 25 m and ⩽ **50 m**	New ticket plus a **candidate** for officer review.
> 50 m	New ticket, no candidate.

Open tickets are those in `reported`, `verified` or `assigned`.

6.3 Extra rules

- **Same person:** one account, same spot (⩽ 25 m), within 24 hours → counts once toward `reporter_count`.
- **Recently fixed:** a ticket in the same group resolved within the last **14 days** within 25 m → flag the new report as a possible reopen, and ask the reporter "Still there?"; don't merge into a resolved ticket.
- **Officer decisions are final for that pair:** if an officer says "Different", that pair is never suggested again. A merge can be split for 10 minutes by anyone allowed to merge, and later by a Ward Officer.

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7. Data retention

Dates count from the event shown. Times are in `rule_config`. The reasoning, law references and exceptions are in `security.md` §6.

Data	How long	Then
Full report photo (⩽ 300 KB) of an **open** ticket	While the ticket is open	
Full report photo of a **resolved** ticket	**90 days** after `resolved_at`	Deleted
Full photo of a **rejected** ticket	**30 days** after rejection	Deleted
Photos flagged `privacy_blur_failed`	**Half** of the above	Deleted
After-photo (crew's fix photo)	90 days (full), thumbnail 12 months	Full deleted; thumbnail deleted at 12 months
Report thumbnail (⩽ 30 KB)	**12 months** after `resolved_at`	Deleted
Link between a report and the citizen's account (`reporter_id`)	**180 days** after `resolved_at`	Link removed (set to empty); the report stays as an anonymous record
Ticket and report records (class, time, location of the hazard, status, scores)	**3 years** after `resolved_at` (**A-08**)	Deleted, or kept only as ward-level aggregates
Device push tokens	Until sign-out, error response or **12 months** of inactivity	Deleted
Staff audit log	**12 months**	Deleted
Staff account data	While active, plus **90 days** after disabling	Deleted
Unsent reports in the app	**7 days**	Person decides: send or discard
Crash and diagnostic logs	Provider default (about 90 days)	Auto-deleted
Citizen "delete my data" request	Completed within **30 days**	Reporter link removed; photos not needed for an open ticket deleted; the ticket itself stays as a public-interest record

Backups: provider backups may keep deleted data for a short period (check the plan's backup window). The privacy notice says so in plain words.

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8. Classification and confidence rules

8.1 How the phone uses the model's confidence

Model's top confidence	What the person sees
≥ 0.70	The class is pre-selected and marked "Our guess".
0.50 – 0.69	The class is shown first and marked "Maybe", **not** pre-selected.
< 0.50, or top class is `none_of_these`	No guess is shown: "We couldn't tell what this is. Pick the closest one."

8.2 The person decides

The stored `class_final` is always what the person confirmed. The model's guess is stored in `class_model`. Neither the phone nor the cloud overrides the person.

8.3 Learning loop

- We log where the person disagreed with the model, along with `model_version`.
- Images used for retraining must be either (a) collected by the team, (b) from open datasets with compatible licences, or (c) from citizens who explicitly opted in. Citizen photos are **not** used for training by default.
- Only officer-verified tickets are used as labelled examples.

8.4 Fallback class set

If by **Day 40** any class has fewer than **300** usable images, or if macro-F1 is below **0.75** at M3, we fall back to three classes: `pothole`, `broken_road`, `other`. Open drains and waterlogging stay available as manual choices under **Other** with a sub-choice, and are scored with their normal C weights. We re-measure and record the decision in `memory.md`.

8.5 Model release gate

A model goes into an app build only if: inference p95 < 150 ms on the reference device (RD-1), size ≤ 5 MB, quality targets met or fallback decision recorded, and `model_version` plus metrics saved with the build.

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9. Tunable parameters (what `rule_config` holds)

Key	Default	Section
`score.w_class` / `w_extent` / `w_conf` / `w_location`	0.30 / 0.25 / 0.15 / 0.30	§1.1
`score.class_weight.*`	See §1.2	§1.2
`score.extent.*`	See §1.3	§1.3
`score.location.*`	See §1.4	§1.4
`score.bonus_per_reporter` / `bonus_cap`	3 / 10	§1.6
`band.critical_min` / `high_min` / `medium_min`	75 / 55 / 35	§1.7
`gps.good_max_m` / `weak_max_m`	30 / 100	§2.1–2.3
`intake.stale_hours`	72	§2.5
`intake.daily_limit`	10	§2.9
`verify.corroboration_reporters` / `window_days`	3 / 14	§3.2
`fastlane.min_conf` / `max_accuracy_m`	0.85 / 20	§3.3
`reopen.confirmations` / `window_days`	2 / 14	§3.6
`sla.*`	See §4 table	§4
`escalate.l1` / `l2` / `l3`	100% / 150% / 200%	§5
`dup.auto_m` / `candidate_m`	25 / 50	§6.2
`dup.recent_resolved_days`	14	§6.3
`retain.*`	See §7	§7
`model.preselect_min` / `maybe_min`	0.70 / 0.50	§8.1
`model.fallback_min_images` / `fallback_deadline_day` / `fallback_f1`	300 / 40 / 0.75	§8.4

Change process: any change to these values is made by the Backend Lead, recorded in the revision log (`loop.md` §6) with a reason, and takes effect without an app release. Changes to *formula shape* (not just numbers) go through a review in `loop.md`.

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10. Document acceptance criteria

This document passes review when it meets the checklist for `rules.md` in `loop.md` §3.